

AI-Assisted Biologics R&D Decision Support

Solving problems, not just showcasing tools

InSynBio integrates **AI Analysis + 6 Proprietary Clinical Databases + Expert Judgment** to provide end-to-end decision support from sequence to report for antibody, bispecific, CAR-T, and vaccine development teams—systematically identifying and mitigating R&D failure risks before entering the wet lab.



CMC Developability Gap

CMC boundaries are not considered during sequence optimization, exposing gaps only at the IND stage, leading to ~30% of candidates failing pre-clinically.

Cost: 6–18 months of wasted development cycles



Delayed Immunogenicity Risk

Inadequate humanization and unscreened T-cell epitopes trigger ADA signals only during animal trials, forcing a complete restart.

Cost: Failed animal trials = million-dollar reinvestment



Format Selection Relies on Experience

Selection of bispecific / ADC / CAR components lacks clinical data support, leading to poor reproducibility and unquantifiable risks when taking the wrong direction.

Cost: Format errors mean redoing the entire pipeline



Our Solution: Intervene before the wet lab—using 25+ AI tools for rapid analysis, 6 clinical databases for format-specific benchmarking, and expert judgment for actionable recommendations. **Not just scores, but directions. Not just tools, but a moat.**

ABOUT THE INSYNBIO TEAM

The core team comes from **Albert Einstein College of Medicine, Columbia University, and Rockefeller University**, with immunology backgrounds and over a decade of practical experience in biologics development. We focus on combining computational AI with deep immunological knowledge, having served multiple Biotech companies and research teams with full computational analysis delivery from antibody humanization to neoantigen vaccines.

🎓 Albert Einstein College of Medicine 🎓 Columbia University 🎓 Rockefeller University 🧬 Immunology Expertise

🔧 10+ Years Biologics Dev 🧑‍💻 AI Engineering

1,142

Clinical Antibody
Reference Sequences

138

Rare ADA
Immune Records

20+

AI Analysis
Tools

9

Complete
Case Studies

100+

ADC Clinical
Projects

3–5 Days

Expert-Reviewed
Delivery

VH/VL Humanization**VHH Humanization****VH→VHH Conversion****CMC Developability****Virtual Affinity Maturation****CDR Redesign****Immunogenicity / ADA****Bispecific Antibodies****ADC Evaluation****CAR-T / NK****Vaccine / Neoantigen**

How does AI truly serve biologics R&D?

Not tools, not experiments — it's decision support

Providing AI tools alone, without clinical data and expert judgment, yields scores that cannot guide decisions. Relying solely on CRO experiments without pre-wet-lab direction means every trial-and-error burns money and time. InSynBio offers a third path: **Using 1,142 clinical sequences as a benchmarking baseline, 25+ tools for cross-validation to eliminate single-tool bias, and immunological expert interpretation to make reports directly actionable.** These three capabilities are indispensable and hard to replicate.

AI Tool Platforms	InSynBio	Traditional CROs
Rosetta · ESMfold · Online Services	AI + Clinical Data + Expert Judgment	Wet Lab Outsourcing
✓ Fast automation	✓ 25+ AI tools running synergistically	✓ Direct wet lab validation
✓ Partially open-source	✓ 6 Proprietary Clinical Databases	✓ GMP · Regulatory support
✗ No clinical format benchmarking data	✓ Format-specific benchmarking (VHH/VH-VL/BsAb)	✗ No early computational risk screening
✗ No Expert Interpretation and Review	✓ Full Review by Immunology Experts	✗ High cost (10–100× computational)
✗ Single tool, no integrated judgment	✓ Actionable optimization advice, not isolated scores	✗ Cycle takes weeks to months
✗ Outputs scores, no actionable advice	✓ 3–5 business days delivery	✗ Cannot identify CMC risks before experiments
✗ No rare ADA / ADC data	✓ 138 rare ADA data records	✗ No clinical sequence benchmarking baseline
✗ No domain background knowledge support	✓ 10+ Years Biologics Dev Background	✗ No computational optimization advice provided

Optimal Path: First use InSynBio to identify key risks and determine directions at the computational layer, then advance high-confidence candidates to wet lab validation—spending the CRO budget on what matters, not on trial-and-error.

From Sequence to Report — 10 Full-Pipeline Services

A single team, one integrated report. Each service can be commissioned independently or combined into an end-to-end project.

► ANTIBODY ENGINEERING



Antibody Humanization

VH/VL Humanization

- CDR Grafting + Vernier Zone Analysis
- 1,142 clinical sequence format benchmarking
- Precise back-mutations + conformational validation

VH/VL mAb



VHH Humanization / Conversion

VHH · VH→VHH

- 29-Item VHH-Specific Checklist
- VH→VHH Structural Conversion Design
- Preserve CDR binding while improving developability

VHH Nanobody



CMC Developability Assessment

Developability · 15 Metrics

- Comprehensive scan of 15 metrics (pI · Hydrophobicity · Charge)
- **Format-Specific:** VH/VL / VHH / Bispecific individual benchmarking
- Shortcoming identification + AI optimization advice

CMC Clinical Benchmark



Virtual Affinity Maturation

VAM · $\Delta\Delta G$ 6-Tool

- EvoEF2 / PRODIGY / FoldX / ThermoMPNN
- AntiFold / ESM-IF1 / MM-GBSA Synergy
- Mutation heatmap + Multi-structure consensus screening

$\Delta\Delta G$ Affinity



CDR Redesign

CDR Redesign · ProteinMPNN

- ProteinMPNN diverse sequence generation
- CDR conformation anchor maintenance, RMSD monitoring
- Shannon polymorphism scoring + Affinity filtering

ProteinMPNN CDR



Immunogenicity and ADA Risk

Immunogenicity · ADA DB

- T-cell epitope screening, 27 HLA-DR coverage
- Benchmarking against 138 **Rare ADA Database** records
- Immunogenicity hotspot localization + Mutation risk reduction

ADA T-cell Epitope

► MODALITY EXTENSION



Bispecific Antibodies

Bispecific · 134+ Molecules

- 134+ Clinical Bispecifics + 84 Linker Library
- Format screening · Pairing stability · CMC mismatch
- 78 IgG-like structural model validation

Bispecific SmartLink



ADC Design & Evaluation

ADC · 100+ Clinical Projects

- 100+ ADC clinical project database
- Linker / Payload / DAR Optimization
- Conjugation site screening + Stability assessment

ADC Linker



CAR-T / NK / M Design

CAR Immunotherapy · 237 Parts

- 237 Selected components, 12-category system architecture
- Hinge / Costimulatory domain rational optimization
- Solid tumor penetration · Safety switch design

CAR-T CAR-NK



Vaccine / Neoantigen Design

Vaccine · Neoantigen · mRNA

- KRAS G12D · EBV and other target cases
- 100+ IEDB validated epitopes, MHC-I/II
- Multi-epitope mRNA assembly + GC optimization

mRNA Neoantigen



De Novo Protein Drug Design

De Novo Protein Drug Design

Research Topic

- RFdiffusion · ProteinMPNN · AlphaFold2 synergistic generation
- Independent of existing scaffolds, exploring novel binding sequence spaces
- De novo design + structure prediction validation + developability pre-assessment

De Novo RFdiffusion

Research · Collaborative Exploration



TCR-Restricted Epitope Immunopotential

TCR-Restricted Epitope Immunopotential

Research Topic

- Binding energy optimization of TCR recognizing pMHC epitopes
- Enhancing T-cell immune response, multi-HLA coverage strategy
- pMHC-TCR interface AI analysis and mutation design

TCR-pMHC Immune Enhancement

Research · Collaborative Exploration

03 · CORE ASSETS

6 Proprietary Databases + 25+ AI Tools

This is an asset portfolio that other AI platforms and CROs cannot replicate—not a simple integration of public tools, but clinically validated proprietary databases and synergistic AI analysis capabilities.

1,142 Premium

Clinical Antibody Sequence Database

Covers VH/VL, VHH, and bispecific formats, classified benchmarking, with clinical status annotations

138 RARE

ADA Immunogenicity Record Database

Curated clinical ADA events, the only proprietary reference baseline for anti-drug antibody risk judgment

100+ 

ADC Clinical Project Database

Full coverage of Linkers, Payloads, DARs, and targets, supporting rational ADC format selection

134+

Bispecific Molecule Database

Clinical bispecific formats · 84 validated Linkers · 78 IgG-like structural models

237 Comprehensive

CAR Component Database

Systematic classification of 12 component categories, supporting CAR-T / NK / Macrophage tri-modality design

100+

IEDB Validated Vaccine Epitopes

27 HLA-DR coverage, MHC-I/II dual screening, supported by clinical tumor neoantigen cases

► AI TOOL MATRIX (25+ TOOLS · 5 IN-HOUSE MODULES, 6 CATEGORIES) ★ In-House Module

STRUCTURE PREDICTION · MOLECULAR DOCKING

ABodyBuilder2AlphaFold2ANARCI / AbNumHADDOCK3

ENERGY CALCULATION · AFFINITY

EvoEF2PRODIGYFoldXThermoMPNNAntiFoldESM-IF1MM/GBSAOpenMM

CMC · DEVELOPABILITY

★ AbEvaluatorTAPAbLangSAP / PSHpl · GRAVY · CDR Liability

IMMUNOGENICITY ASSESSMENT ★ IN-HOUSE

IEDB API★ MHC-II Analyzer★ ADA Risk Scorer★ Surface Immunogenicity Detection

VACCINE EPITOPE DESIGN ★ IN-HOUSE

★ Multi-Epitope Assembler★ Neoantigen ScannerMHCflurry 2.0★ Heteroclitic Designer★ Codon Optimizer★ Epitope Prioritization Scoring

CDR DESIGN · DE NOVO GENERATION

ProteinMPNNESM-2RFdiffusionAfDesignBindCraft

04 · REAL-WORLD CASES

9 Complete Cases — Covering All Service Types

Includes published methodological cases and several anonymized client projects. Reports available upon request (under NDA).

ANTIBODY HUMANIZATION

Case Demo

muMAb4D5 → Herceptin Humanization Reproduction

4 Back-mutations · 15/15 CMC ✓

A systematic, data-driven re-humanization of the world's most clinically validated antibody using InSynBio AbEngineCore V4.5.

→ Avoid wrong frameworks entering synthesis, saving 2–4 weeks of rework and re-commissioning costs

VH→VHH CONVERSION

Case Demo

muMAb4D5 VH→VHH Nanobody Design

HER2 Affinity Retained · VGRW_SR_R2

Ready-to-synthesize HER2-targeting VHH nanobody. All 3 CDRs preserved at 0.054 Å RMSD. Molecular docking confirms engagement.

→ Structure and function combined: affinity retained + thermal stability improved, directly advancing pre-IND optimization

CMC DEVELOPABILITY

Case Demo

muMAb4D5 CMC Assessment — Clinical- Grade

All 15 Parameters Passed · AbRef- 458

Benchmarked against 458 clinical antibodies — and surpassing Herceptin® on immunogenicity safety.

→ Pre-clinically confirm candidate developability, reducing Phase I immunogenicity risks

VIRTUAL AFFINITY MATURATION

Case Demo

VGRW-SR-R2 Anti-HER2 VHH VAM

Exhaustive interface scanning · 247 mutations

Evaluated across 5 independent dimensions, uncovering an epistatic synergy pair.

→ Computational pre-screening replaces random mutation library screening, saving 60–80% in synthesis and expression costs

VIRTUAL AFFINITY MATURATION

Client Project

Fentanyl Hapten VAM (Hapten-VAM)

ΔΔG Consensus –1.8 kcal/mol

Computational optimization of a clinical-grade anti-fentanyl antibody: multi-stage ΔΔG screening from 100+ candidate mutations.

→ Identified a synergistic pair, saving 60–80% in synthesis and expression costs

CDR REDESIGN

Case Demo

Structure-Guided CDR Redesign & Re- maturation

De novo CDR2/CDR3 sequence reconstruction

Starting from the SR-R2 champion VHH, we apply structure-constrained generative AI to expand patent space.

→ 87 diverse candidate sequences, avoiding sequence diversity bottlenecks and patent infringement

BISPECIFIC ANTIBODIES

Case Demo

Faricimab VH/VL Pairing Stability Audit

InSynBio VPSA reveals intrinsic pairing selectivity

KiH+CrossFab co-expression solved chain mispairing — but do Faricimab's two Fv interfaces have intrinsic pairing selectivity?

→ Rationally locked in the optimal pairing, avoiding full-coverage blind testing in wet labs

EPITOPE ANALYSIS

Client Project

PD-L1 Dual-Clone Epitope Structural Analysis

Lateral (non-blocking) vs. frontal (blocking)

AF2-Multimer structural modeling precisely maps PD-L1 binding epitopes of two client antibodies.

→ Exposed shortcomings 3–6 months early, completing fixes before IND, avoiding passive pre-clinical elimination

CAR-M DESIGN

Case Demo

Anti-CIDRα1 Dual- Binder CAR- Macrophage

High-Parasitemia *P. falciparum* Malaria

A rationally designed CAR-M construct with every decision backed by the ACTES 11-layer decision cascade.

→ Novel CAR-M modality for infectious diseases, expanding beyond traditional oncology applications

Full reports available upon request under NDA · Client Project data has been anonymized

05 · COLLABORATION MODEL

Starting with One Email

No formal documents required upfront. Let's discuss the project background first, and we will determine the scope of analysis. **All projects sign an NDA before starting**—your sequences, targets, and data are legally protected and kept strictly confidential, never disclosed to third parties, and never used for training or public release.

Project-Based Service

- Single commission, clear scope and deliverables

Resident AI Consultant

- 3–5 business days Expert-reviewed report
- NDA protection · Milestone billing
- Balance waived if objectives are not met

- Monthly / quarterly retainer, continuous decision support
- Participate in R&D meetings, real-time AI analysis response
- Multi-project parallel support, flexible allocation
- Suitable for Biotech teams with continuous R&D pipelines



Free Initial Assessment

Explain the project via email, feedback same or next day

Sign NDA

Sequences, targets, and data are legally protected, not leaked, not used for training

Confirm Scope

Analysis format, tool selection, deliverable agreement

AI + Expert Review

25+ Tools running, Expert full-process review of the report

3-5 Days Delivery

Full report + recommendations · Balance waived if objectives are not met



PREFERRED · QUICK RESPONSE

contact@insynbio.com

Email us with your inquiry



FORMAL PROJECT / NDA

contact@insynbio.com

Formal commission or NDA signing



ONLINE PLATFORM

www.insynbio.com

Service details · Online project submission

This document is based on computational analysis and public Clinical Data, does not constitute clinical decision advice, and is for R&D reference only. © 2026 InSynBio